Task 1: Create an IAM role for read-only S3 access and a user named s3viewer

Objective: Allow the user to view S3 data without permission to upload, delete or modify objects.

Step 1: Create a read-only policy

1. Sign in to the AWS Management Console.
2. Search for IAM.
3. Open Policies → Create policy.
4. Select the JSON editor and paste this policy.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadOnlyS3Access",
      "Effect": "Allow",
      "Action": [
        "s3:ListAllMyBuckets",
        "s3:ListBucket",
        "s3:GetObject"
      ],
      "Resource": [
        "*",
        "arn:aws:s3:::*",
        "arn:aws:s3:::*/*"
      ]
    }
  ]
}
```

5. Choose Next, name the policy S3ReadOnlyPolicy, and create it.

Step 2: Create the IAM role

1. Open IAM → Roles → Create role.

2. Choose AWS account as the trusted entity type.
3. Select your own account and create the role.
4. Name it S3ReadOnlyRole.
5. Attach S3ReadOnlyPolicy and create the role.

Step 3: Create the user
1. Go to IAM → Users → Create user.
2. Enter the username s3viewer.
3. Complete the creation process.
4. Configure the user to assume S3ReadOnlyRole using the appropriate sign-in method and permissions.

Task 2: Configure S3 server-side encryption with automatic key rotation

Objective: Protect images stored in the bucket my-zomato-images using AWS Key Management Service (KMS).

Step 1: Create the KMS key
1. Search for KMS in the AWS Console.
2. Open Customer managed keys → Create key.
3. Select Symmetric and the encrypt/decrypt key usage.
4. Choose the appropriate key type and proceed.
5. Set the alias to alias/zomato-images-key.
6. Configure key administrators and authorized users.
7. Create the key.

Step 2: Enable automatic key rotation
1. Open the newly created customer-managed KMS key.
2. Find the Key rotation section or the rotation action in the key details.
3. Enable automatic rotation.
4. Verify that the key's rotation status is enabled.

Step 3: Create and configure the S3 bucket

1. Open Amazon S3.
2. Select Create bucket.
3. Enter my-zomato-images if that name is available globally.
4. Select your required AWS Region.
5. Keep Block all public access enabled.
6. Create the bucket.
7. Open the bucket and select Properties → Default encryption.
8. Choose Server-side encryption with AWS KMS keys (SSE-KMS).
9. Select your customer-managed key, alias/zomato-images-key, and save the configuration.

Step 4: Verify encryption
Upload a test image, open its object details and inspect its server-side encryption settings. Confirm that the object uses SSE-KMS and that the expected KMS key is selected.

Task 3: Enable CloudWatch monitoring and logging for an EC2 web app

Objective: Monitor the performance of an EC2 instance hosting a small product-list website and verify that logs are being collected.

Step 1: Check EC2 metrics
1. Open EC2 → Instances.
2. Select your running web server instance.
3. Open the Monitoring tab.
4. Review metrics such as CPU utilization, network traffic and status checks.

Step 2: Create a log group
1. Open CloudWatch → Logs → Log groups.
2. Choose Create log group.
3. Enter /ec2/flipkart-webapp.
4. Create the log group and configure a suitable retention period.

Step 3: Configure log collection
1. Give the EC2 instance an IAM role that includes the permissions required to publish logs to CloudWatch, such as those in the AWS-managed CloudWatchAgentServerPolicy.
2. Install the CloudWatch agent on the instance using the installation instructions for its operating system.
3. Configure the agent to collect the web server's access and error logs. For example, Apache logs on a typical Ubuntu installation are often located under /var/log/apache2/.
4. Configure the agent to send those logs to /ec2/flipkart-webapp.
5. Start the agent and check its status.

Step 4: Verify logs
Open CloudWatch → Logs → Log groups → /ec2/flipkart-webapp and inspect the log streams.

Generate a harmless test web request to your own application, then refresh the log stream. Confirm that the request appears in the access log and that the log timestamps are current.

Task 4: Create a CloudWatch CPU alarm that sends an email

Objective: Receive an email notification whenever EC2 CPU utilization goes above 70%.

Step 1: Create an SNS topic
1. Open the Amazon SNS Console.
2. Select Topics → Create topic.
3. Choose Standard.
4. Name it EC2-CPU-Alert.
5. Create the topic.
6. Open the topic and choose Create subscription.
7. Select Email as the protocol and enter your email address.
8. Create the subscription and confirm it using the link received by email.

Step 2: Create the CloudWatch alarm
1. Open CloudWatch → Alarms → All alarms.
2. Choose Create alarm → Select metric.
3. Select EC2 → Per-Instance Metrics.
4. Find and select the CPUUtilization metric for your instance.
5. Configure the following settings.



6. Configure the alarm to trigger when the selected metric breaches the threshold. For example, one evaluation period can be used for a basic lab.\
7. Select the SNS topic and create the alarm.

Step 3: Test the alarm
- Open the alarm details and verify that its state is visible.
- If you want to test it, generate controlled CPU load on your own lab instance, while monitoring resource usage.
- Confirm that the CPU metric crosses the threshold and the alarm enters the ALARM state.
- Check that the email notification arrives.

Task 5: Generate a least-privilege S3 upload/delete policy

Objective: Allow a test user to upload and delete objects in my-insta-uploads, without granting permissions to read objects or administer the bucket.

Step 1: Copy the IAM policy JSON

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "UploadAndDeleteObjectsOnly",
      "Effect": "Allow",
      "Action": [
        "s3:PutObject",
        "s3:DeleteObject"
      ],
      "Resource": "arn:aws:s3:::my-insta-uploads/*"
    }
  ]
}
```

Step 2: Attach the policy to a test user

1. Open IAM → Policies → Create policy.
2. Select the JSON editor and paste the policy above.
3. Name it InstaUploadsOnlyPolicy.
4. Create the policy.
5. Open IAM → Users and select your test user.
6. Choose Add permissions → Attach policies directly.
7. Select InstaUploadsOnlyPolicy and add the permissions.

Step 3: Verify the permissions

Action	Expected result

Upload an object to my-insta-uploads	Allowed
Delete an object in my-insta-uploads	Allowed
Download an object	Not allowed by this policy
Change bucket settings	Not allowed by this policy
Upload or delete objects in another bucket	Not allowed by this policy